HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines on **AWS Lambda** to process and transform approximately **3,500 events per second**, replacing a less efficient batch processing system.
- Developed and deployed **LLM-based applications** and **AI agents** using frameworks like **LangGraph** to automate complex workflow orchestration for enterprise stakeholders.
- Contributed to the development of **AI governance frameworks** and **responsible AI practices**, ensuring adherence to data privacy, transparency, and explainability standards.
- Engineered **Java Spring Boot** REST APIs for cross-service data exchange, incorporating robust error handling and circuit-breaker patterns to enhance system resilience.
- Explored, prepared, and analysed **structured and unstructured data** using **Python** (pandas, NumPy) to identify key patterns and trends for predictive modelling.
- Supported the transition of AI use cases from proof-of-concept to production, including rigorous **testing**, **performance monitoring**, and **continuous improvement** cycles.
- Containerized **Java** and **Python** services using **Docker** and deployed them to a **Kubernetes** cluster (EKS), managing configuration and resource allocation for predictable rollouts.

SKILLS

AI/ML Development – Machine Learning, Applied AI, LLMs, Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, LangGraph, Microsoft Copilot Studio

Programming Languages – Python, Java, SQL, R

Data Science & Analytics – Data Modelling, Statistical Modelling, Predictive Analytics, Exploratory Data Analysis, Feature Engineering, Model Development, Model Evaluation, Data Preparation, A/B Testing, Time-Series Analysis

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, CI/CD, Jenkins, Git, Linux

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, LangGraph, FastAPI

Databases & Data Warehousing – PostgreSQL, MySQL, Oracle, Data Warehousing, ETL, Power Query

BI & Visualization – Power BI, Tableau, DAX, R/RStudio

PROJECTS

Intelligent Automation for Financial Workflows

- Developed an **AI agent** using **LangGraph** to automate reconciliation workflows for financial transactions, reducing manual processing time by an estimated **30%**.
- Implemented a **Python**-based data pipeline to extract and prepare transaction data from disparate sources, ensuring data quality for downstream **machine learning model** training.
- Designed and deployed the solution on **AWS**, incorporating **Docker** containers for consistent environments and **CI/CD** pipelines for automated testing and deployment.

Predictive Analytics for Operational Efficiency

- Built and evaluated a **scikit-learn**-based predictive model to forecast potential system failures, achieving an **AUC of 0.84** on a dataset of historical operational logs.
- Engineered relevant features from system logs and performance metrics using **pandas** and **SQL**, contributing to the model's accuracy.
- Created interactive **Power BI** dashboards to visualize model predictions and key performance indicators for business stakeholders, facilitating proactive maintenance planning.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)